

Measurable Functions and the Lebesgue Integral

II

Anqiao Ouyang

March 2024 · Revised September 2026

Introduction

Article I set up the language of measure spaces $(X, \mathcal{F}, \mu)$ and, through outer measures and the Carathéodory construction, produced the Lebesgue measure space $(\mathbb{R}^n, \mathcal{L}, m)$. The task of this article is to build an integral on top of a measure.

The Riemann integral we know starts from a partition of the **domain**: cut $[a, b]$ into small intervals and approximate f on each of them by one of its values. Lebesgue's idea is to partition the **range** instead: split the values of f into layers

$$y_0 < y_1 < \cdots < y_N,$$

and use the measure to “weigh” the set belonging to each layer,

$$\{x : y_{k-1} \leq f(x) < y_k\}.$$

For this to make sense these sets must be measurable, which leads naturally to the notion of a **measurable function**.

The plan is as follows. Section 1 reviews the Riemann integral and explains its limitations with respect to limits. Section 2 treats measurable functions, the notion of “almost everywhere”, approximation by simple functions, and Lusin's theorem. Section 3 defines the Lebesgue integral successively for non-negative simple functions, non-negative measurable functions and general measurable functions, and establishes its basic properties. Section 4 proves the Monotone Convergence Theorem, Fatou's lemma and the Dominated Convergence Theorem. Section 5 shows how the Lebesgue integral extends the Riemann integral and how it relates to improper integrals.

Normed spaces, the L^p spaces and the inequalities of Hölder and Minkowski are deferred to Article III; uniform integrability and the Vitali convergence theorem are treated in the supplementary article on uniform integrability.

Conventions. Throughout, $(X, \mathcal{F}, \mu)$ denotes a measure space. On $\mathbb{R}^n$ the measure is Lebesgue measure m unless stated otherwise, and we also write $\int_E f(x) dx$ for $\int_E f dm$. The set $\{x \in X :$

$f(x) > a$ is abbreviated $\{f > a\}$. For $A \subseteq X$, $\mathbf{1}_A$ denotes its indicator function: $\mathbf{1}_A(x) = 1$ if $x \in A$ and 0 otherwise.

1 The Riemann Integral and Its Limitations

1.1 Partitions and Riemann Sums

DEFINITION 1.1 *Partitions and Riemann sums*

Let $a < b$. A **partition** of $[a, b]$ is a finite set

$$P = \{x_0, x_1, \dots, x_n\}, \quad a = x_0 < x_1 < \dots < x_n = b.$$

Write $\Delta x_i = x_i - x_{i-1}$ ($1 \leq i \leq n$) and call

$$\|P\| = \max_{1 \leq i \leq n} \Delta x_i$$

the **mesh** of P . A choice of points $\xi_i \in [x_{i-1}, x_i]$ gives a **tag** $\xi = (\xi_1, \dots, \xi_n)$ of P . For $f : [a, b] \rightarrow \mathbb{R}$,

$$S(f, P, \xi) = \sum_{i=1}^n f(\xi_i) \Delta x_i$$

is the corresponding **Riemann sum**.

DEFINITION 1.2 *Riemann integrability*

A function $f : [a, b] \rightarrow \mathbb{R}$ is **Riemann integrable** on $[a, b]$ if there is a real number I such that for every $\varepsilon > 0$ there exists $\delta > 0$ with

$$|S(f, P, \xi) - I| < \varepsilon$$

for **every** partition P with $\|P\| < \delta$ and **every** tag ξ of P . Such an I is unique; it is denoted $\int_a^b f(x) dx$, and we write $I = \lim_{\|P\| \rightarrow 0} S(f, P, \xi)$. The set of Riemann integrable functions on $[a, b]$ is denoted $\mathcal{R}[a, b]$.

REMARK 1.3

The limit is taken as the mesh $\|P\| \rightarrow 0$, uniformly in the choice of tags. Letting only the number of partition points $n \rightarrow \infty$ is not enough. For instance, take $f(x) = x$ on $[0, 1]$ and

$$P_n = \left\{0, \frac{1}{2n}, \frac{2}{2n}, \dots, \frac{n}{2n}, 1\right\}.$$

The number of points tends to infinity, but $[\frac{1}{2}, 1]$ is never subdivided. Choosing $\xi = \frac{1}{2}$ or $\xi = 1$ on that interval, the Riemann sums tend to $\frac{1}{8} + \frac{1}{4} = \frac{3}{8}$ and $\frac{1}{8} + \frac{1}{2} = \frac{5}{8}$ respectively, whereas the integral equals $\frac{1}{2}$.

PROPOSITION 1.4

If $f \in \mathcal{R}[a, b]$, then f is bounded on $[a, b]$.

Proof. Take $\varepsilon = 1$ with its δ and fix a partition P with $\|P\| < \delta$. If f were unbounded, it would be unbounded on some subinterval $[x_{j-1}, x_j]$. Keeping the tags on the other subintervals fixed and varying only ξ_j changes the Riemann sum by $f(\xi_j)\Delta x_j$, which can be made arbitrarily large, contradicting $|S(f, P, \xi) - I| < 1$ for all tags. $\square$

1.2 Darboux Sums**DEFINITION 1.5** *Darboux sums*

Let f be bounded on $[a, b]$ and let P be a partition. Put

$$M_i = \sup_{[x_{i-1}, x_i]} f, \quad m_i = \inf_{[x_{i-1}, x_i]} f.$$

The numbers

$$U(f, P) = \sum_{i=1}^n M_i \Delta x_i, \quad L(f, P) = \sum_{i=1}^n m_i \Delta x_i$$

are the **upper** and **lower Darboux sums** of f with respect to P .

Clearly $L(f, P) \leq S(f, P, \xi) \leq U(f, P)$ for every tag ξ , and

$$U(f, P) = \sup_{\xi} S(f, P, \xi), \quad L(f, P) = \inf_{\xi} S(f, P, \xi).$$

LEMMA 1.6

Let f be bounded.

- (i) If $Q \supseteq P$ (we say Q **refines** P), then $L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P)$.
- (ii) For any two partitions P, P' , $L(f, P) \leq U(f, P')$.

Proof. (i) It suffices to add one point y inside some $[x_{j-1}, x_j]$. The suprema of f over $[x_{j-1}, y]$ and $[y, x_j]$ are at most M_j , so the upper sum does not increase; likewise the lower sum does not decrease. (ii) By (i), $L(f, P) \leq L(f, P \cup P') \leq U(f, P \cup P') \leq U(f, P')$. $\square$

LEMMA 1.7

Let $|f| \leq M$ and let Q be a partition with N interior points (points other than a, b). Then for every partition P ,

$$U(f, P) - L(f, P) \leq U(f, Q) - L(f, Q) + 4MN\|P\|.$$

Proof. Insert the interior points of Q into P one at a time to obtain $P \cup Q$. Each inserted point y

lies in some subinterval J of the current partition, with $|J| \leq \|P\|$, and the upper sum changes by

$$0 \leq \sup_J f \cdot |J| - \left(\sup_{J'} f \cdot |J'| + \sup_{J''} f \cdot |J''| \right) \leq 2M|J| \leq 2M\|P\|,$$

where J', J'' are the two pieces of J . Hence $U(f, P) \leq U(f, P \cup Q) + 2MN\|P\|$ and similarly $L(f, P) \geq L(f, P \cup Q) - 2MN\|P\|$. By Lemma 1.6(i), $U(f, P \cup Q) - L(f, P \cup Q) \leq U(f, Q) - L(f, Q)$. Combining the three inequalities gives the claim. $\square$

THEOREM 1.8 *Darboux criterion*

Let f be bounded on $[a, b]$. The following are equivalent:

- (i) $f \in \mathcal{R}[a, b]$;
- (ii) for every $\varepsilon > 0$ there is a partition Q with $U(f, Q) - L(f, Q) < \varepsilon$.

In this case

$$\int_a^b f(x) dx = \inf_P U(f, P) = \sup_P L(f, P),$$

and for every sequence of partitions with $\|P_k\| \rightarrow 0$ both $U(f, P_k)$ and $L(f, P_k)$ converge to $\int_a^b f(x) dx$.

Proof. (i) $\Rightarrow$ (ii): Let $I = \int_a^b f$. Given $\varepsilon > 0$ take δ as in the definition. If $\|P\| < \delta$, all Riemann sums lie in $(I - \varepsilon, I + \varepsilon)$; taking the supremum and infimum over tags gives

$$I - \varepsilon \leq L(f, P) \leq U(f, P) \leq I + \varepsilon.$$

This yields (ii), as well as $U(f, P_k), L(f, P_k) \rightarrow I$ whenever $\|P_k\| \rightarrow 0$. Together with Lemma 1.6(ii) it also gives $\sup_P L = \inf_P U = I$.

(ii) $\Rightarrow$ (i): Let $I_* = \sup_P L(f, P)$ and $I^* = \inf_P U(f, P)$. By Lemma 1.6(ii), $I_* \leq I^*$; by (ii), $I^* - I_* < \varepsilon$ for every ε , so they are equal; call the common value I . Let $|f| \leq M$. Given $\varepsilon > 0$, choose Q with $U(f, Q) - L(f, Q) < \varepsilon$, let N be its number of interior points and put $\delta = \varepsilon/(4MN + 1)$. By Lemma 1.7, $U(f, P) - L(f, P) < 2\varepsilon$ whenever $\|P\| < \delta$. Since both I and $S(f, P, \xi)$ lie in $[L(f, P), U(f, P)]$,

$$|S(f, P, \xi) - I| \leq U(f, P) - L(f, P) < 2\varepsilon.$$

Hence $f \in \mathcal{R}[a, b]$ with integral I . $\square$

EXAMPLE 1.9

Every $f \in C[a, b]$ is Riemann integrable: f is uniformly continuous on the compact interval, so for $\varepsilon > 0$ there is $\delta > 0$ with $|f(x) - f(y)| < \varepsilon$ whenever $|x - y| < \delta$. If $\|P\| < \delta$, then $M_i - m_i \leq \varepsilon$ for all i , hence $U(f, P) - L(f, P) \leq \varepsilon(b - a)$.

1.3 Limitations of the Riemann Integral

The Riemann integral works well for continuous functions, but its shortcomings appear as soon as limits are involved.

(1) Not closed under limits.

EXAMPLE 1.10

Enumerate $[0, 1] \cap \mathbb{Q}$ as $r_1, r_2, \dots$ and let

$$f_n = \mathbf{1}_{\{r_1, \dots, r_n\}}.$$

For every partition P , each r_k belongs to at most two subintervals, so $L(f_n, P) = 0$ and $U(f_n, P) \leq 2n\|P\|$. By Theorem 1.8, $f_n \in \mathcal{R}[0, 1]$ and $\int_0^1 f_n = 0$. But f_n increases pointwise to the Dirichlet function

$$D = \mathbf{1}_{\mathbb{Q} \cap [0, 1]},$$

and since every subinterval contains both rational and irrational points, $U(D, P) = 1$ and $L(D, P) = 0$ for all P ; thus $D \notin \mathcal{R}[0, 1]$.

So even a uniformly bounded, increasing sequence of Riemann integrable functions may have a pointwise limit that is not Riemann integrable.

(2) Interchanging limits and integrals. In the Riemann theory the standard sufficient condition is uniform convergence.

PROPOSITION 1.11

Let $f_n \in \mathcal{R}[a, b]$ and $f_n \rightarrow f$ uniformly on $[a, b]$. Then $f \in \mathcal{R}[a, b]$ and

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx.$$

Proof. Given $\varepsilon > 0$, choose n with $\sup_{[a, b]} |f_n - f| < \varepsilon$. Then f is bounded, and for every partition P ,

$$U(f, P) - L(f, P) \leq U(f_n, P) - L(f_n, P) + 2\varepsilon(b - a).$$

Choosing P so that the first term on the right is less than ε , Theorem 1.8 gives $f \in \mathcal{R}[a, b]$. Moreover $|S(f_n, P, \xi) - S(f, P, \xi)| \leq \varepsilon(b - a)$ for all P, ξ ; letting $\|P\| \rightarrow 0$ yields $|\int_a^b f_n - \int_a^b f| \leq \varepsilon(b - a)$. $\square$

REMARK 1.12

Uniform convergence is **sufficient but not necessary**. For example, $f_n(x) = x^n$ converges on $[0, 1]$ to $f = \mathbf{1}_{\{1\}}$, not uniformly, yet

$$\int_0^1 x^n dx = \frac{1}{n+1} \rightarrow 0 = \int_0^1 f(x) dx.$$

The classical theorem of Arzelà states: if $f_n, f \in \mathcal{R}[a, b]$ are uniformly bounded and $f_n \rightarrow f$ pointwise, then the integrals converge. Within the Riemann framework this is quite laborious to prove; in the Lebesgue theory it is an immediate consequence of the Dominated Convergence Theorem (Remark 5.4). The real difficulty of the Riemann theory is that one has to **know in advance** that the limit f is Riemann integrable, and Example 1.10 shows that this cannot be deduced from the integrability of the f_n .

(3) Incompleteness. For $f, g \in \mathcal{R}[a, b]$ put

$$d_1(f, g) = \int_a^b |f(x) - g(x)| dx.$$

(Theorem 1.8 easily shows $|f - g| \in \mathcal{R}[a, b]$.) The function d_1 is symmetric and satisfies the triangle inequality, but $d_1(f, g) = 0$ does not imply $f = g$: changing a function at one point does not change its integral. Thus d_1 is only a **pseudometric**; to obtain a metric space one must identify functions with $d_1(f, g) = 0$. Even so, the resulting space is not complete.

EXAMPLE 1.13

On $[0, 1]$ let $f_n(x) = \min\{n, x^{-1/2}\}$ (with $f_n(0) = n$). Each f_n is continuous and

$$\int_0^1 f_n(x) dx = n \cdot \frac{1}{n^2} + \int_{1/n^2}^1 x^{-1/2} dx = 2 - \frac{1}{n}.$$

For $m > n$ we have $f_m \geq f_n$, hence

$$d_1(f_m, f_n) = \int_0^1 (f_m - f_n) dx = \frac{1}{n} - \frac{1}{m} < \frac{1}{n},$$

so (f_n) is a Cauchy sequence.

Suppose $g \in \mathcal{R}[0, 1]$ satisfied $d_1(f_n, g) \rightarrow 0$. By Proposition 1.4 there is M with $|g| \leq M$. Put $c = (M + 1)^{-2}$. For $n \geq M + 1$ and $x \in [0, c]$ we have $f_n(x) \geq M + 1$, so $|f_n - g| \geq 1$ there and

$$d_1(f_n, g) \geq c > 0 \quad (n \geq M + 1),$$

a contradiction. Hence this Cauchy sequence has no limit in $\mathcal{R}[0, 1]$.

Its “limit” ought to be the unbounded function $x^{-1/2}$, which is not in $\mathcal{R}[0, 1]$. Completing $(\mathcal{R}[a, b], d_1)$ produces exactly the space L^1 studied in Article III.

(4) Unbounded functions and unbounded intervals. The Riemann integral is defined only for bounded functions on bounded intervals. Integrals such as $\int_0^1 x^{-1/2} dx$ or $\int_0^\infty e^{-x} dx$ require an additional limiting procedure (improper integrals), and such integrals are not well behaved (see §5.2).

1.4 Lebesgue’s Idea

Let $f \geq 0$ be bounded. Instead of partitioning the domain, partition the range: choose $0 = y_0 < y_1 < \cdots < y_N$ covering the values of f , put

$$E_k = \{y_{k-1} \leq f < y_k\},$$

and approximate “the integral of f ” from below by

$$\sum_{k=1}^N y_{k-1} \mu(E_k).$$

Such a sum depends only on the measures of the layers, not on whether those sets are made of intervals. It does require every E_k to be measurable, and the next section starts from this requirement.

We should also dispel a common misconception: the Lebesgue integral *is* an extension of the (proper) Riemann integral. Every Riemann integrable function is Lebesgue integrable with the same

integral (Theorem 5.1). The only genuine incompatibility concerns certain conditionally convergent **improper** integrals (§5.2).

2 Measurable Functions

2.1 Definition

We use the **extended real line** $\overline{\mathbb{R}} = [-\infty, +\infty]$ with the usual order and the conventions $a + (+\infty) = +\infty$ for $a > -\infty$, $a \cdot (+\infty) = +\infty$ for $a \in (0, +\infty]$, and

$$0 \cdot (\pm\infty) = 0.$$

The expression $(+\infty) - (+\infty)$ is undefined.

DEFINITION 2.1 *Measurable function*

Let $(X, \mathcal{F})$ be a measurable space and $E \in \mathcal{F}$. A function $f : E \rightarrow \overline{\mathbb{R}}$ is **$\mathcal{F}$ -measurable** (or simply **measurable**) if

$$\{x \in E : f(x) > a\} \in \mathcal{F} \quad \text{for every } a \in \mathbb{R}.$$

If $(X, \mathcal{F}) = (\mathbb{R}^n, \mathcal{L})$, f is called **Lebesgue measurable**; if $\mathcal{F} = \mathcal{B}(\mathbb{R}^n)$, it is called **Borel measurable**.

Since $\mathcal{B}(\mathbb{R}^n) \subseteq \mathcal{L}(\mathbb{R}^n)$ (Article I, §4.5), every Borel measurable function is Lebesgue measurable.

PROPOSITION 2.2

For $f : E \rightarrow \overline{\mathbb{R}}$ the following are equivalent:

- (i) $\{f > a\} \in \mathcal{F}$ for all $a \in \mathbb{R}$;
- (ii) $\{f \geq a\} \in \mathcal{F}$ for all $a \in \mathbb{R}$;
- (iii) $\{f < a\} \in \mathcal{F}$ for all $a \in \mathbb{R}$;
- (iv) $\{f \leq a\} \in \mathcal{F}$ for all $a \in \mathbb{R}$.

In this case $\{f = +\infty\}$, $\{f = -\infty\}$ and $\{f = a\}$ ($a \in \mathbb{R}$) belong to $\mathcal{F}$.

Proof. Since a σ -algebra is closed under countable intersections and complements,

$$\{f \geq a\} = \bigcap_{k=1}^{\infty} \left\{f > a - \frac{1}{k}\right\}, \quad \{f < a\} = E \setminus \{f \geq a\}, \quad \{f \leq a\} = \bigcap_{k=1}^{\infty} \left\{f < a + \frac{1}{k}\right\},$$

and $\{f > a\} = E \setminus \{f \leq a\}$ give (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) $\Rightarrow$ (i). Finally,

$$\{f = +\infty\} = \bigcap_k \{f > k\}, \quad \{f = -\infty\} = \bigcap_k \{f < -k\}, \quad \{f = a\} = \{f \geq a\} \cap \{f \leq a\}.$$

□

PROPOSITION 2.3

A real-valued function $f : E \rightarrow \mathbb{R}$ is measurable if and only if $f^{-1}(B) \in \mathcal{F}$ for every Borel set $B \subseteq \mathbb{R}$.

Proof. Sufficiency is clear (take $B = (a, \infty)$). For necessity, let

$$\mathcal{A} = \{B \subseteq \mathbb{R} : f^{-1}(B) \in \mathcal{F}\}.$$

Since $f^{-1}(\mathbb{R} \setminus B) = E \setminus f^{-1}(B)$ and $f^{-1}(\bigcup_k B_k) = \bigcup_k f^{-1}(B_k)$, $\mathcal{A}$ is a σ -algebra. It contains every (a, ∞) , hence every

$$(a, b) = (a, \infty) \setminus \bigcap_k \left(b - \frac{1}{k}, \infty\right),$$

that is, every open interval. By Article I, §2.5, the open intervals generate $\mathcal{B}(\mathbb{R})$, so $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{A}$. $\square$

EXAMPLE 2.4

- (i) $\mathbf{1}_A$ is measurable if and only if $A \in \mathcal{F}$: the set $\{\mathbf{1}_A > a\}$ is X , A or $\emptyset$.
- (ii) A continuous $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is Borel measurable: $\{f > a\} = f^{-1}((a, \infty))$ is open.
- (iii) A monotone $f : \mathbb{R} \rightarrow \mathbb{R}$ is Borel measurable: $\{f > a\}$ is an interval (possibly empty or unbounded).

2.2 Operations on Measurable Functions**PROPOSITION 2.5**

Let $f, g : E \rightarrow \mathbb{R}$ be measurable, $c \in \mathbb{R}$, and let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be Borel measurable (for example continuous). Then

$$f + g, \quad cf, \quad fg, \quad \max\{f, g\}, \quad \min\{f, g\}, \quad |f|, \quad \varphi \circ f$$

are measurable.

Proof. If $f(x) + g(x) > a$, then $f(x) > a - g(x)$, and there is a rational q with $a - g(x) < q < f(x)$. Hence

$$\{f + g > a\} = \bigcup_{q \in \mathbb{Q}} (\{f > q\} \cap \{g > a - q\}) \in \mathcal{F}.$$

For $c > 0$, $\{cf > a\} = \{f > a/c\}$; for $c < 0$, $\{cf > a\} = \{f < a/c\}$; for $c = 0$, cf is constant. For $\varphi \circ f$, Proposition 2.3 gives

$$\{\varphi \circ f > a\} = f^{-1}(\varphi^{-1}((a, \infty))) \in \mathcal{F}.$$

In particular f^2 is measurable, and therefore so is

$$fg = \frac{1}{4}((f + g)^2 - (f - g)^2).$$

Finally $\{\max\{f, g\} > a\} = \{f > a\} \cup \{g > a\}$, $\{\min\{f, g\} > a\} = \{f > a\} \cap \{g > a\}$, and $|f| = \max\{f, -f\}$. $\square$

REMARK 2.6

The order of composition matters: if φ is merely Lebesgue measurable and f is continuous, $\varphi \circ f$ need not be Lebesgue measurable, because the preimage of a Lebesgue measurable set under a continuous map need not be Lebesgue measurable. This is why φ is required to be Borel measurable.

PROPOSITION 2.7

Let $f_n : E \rightarrow \overline{\mathbb{R}}$ ($n \geq 1$) be measurable. Then

$$\sup_n f_n, \quad \inf_n f_n, \quad \limsup_{n \rightarrow \infty} f_n, \quad \liminf_{n \rightarrow \infty} f_n$$

are measurable; the set

$$C = \left\{ x \in E : \lim_{n \rightarrow \infty} f_n(x) \text{ exists in } \overline{\mathbb{R}} \right\}$$

belongs to $\mathcal{F}$; and if $f_n \rightarrow f$ pointwise on E , then f is measurable.

Proof. $\{\sup_n f_n > a\} = \bigcup_n \{f_n > a\}$ and $\inf_n f_n = -\sup_n (-f_n)$. Hence

$$\limsup_{n \rightarrow \infty} f_n = \inf_{k \geq 1} \sup_{n \geq k} f_n, \quad \liminf_{n \rightarrow \infty} f_n = \sup_{k \geq 1} \inf_{n \geq k} f_n$$

are measurable. Moreover $\liminf_{n \rightarrow \infty} f_n(x) < \limsup_{n \rightarrow \infty} f_n(x)$ if and only if some rational q lies strictly between them, so

$$E \setminus C = \bigcup_{q \in \mathbb{Q}} \left(\{ \liminf_n f_n < q \} \cap \{ \limsup_n f_n > q \} \right) \in \mathcal{F}.$$

If $f_n \rightarrow f$ pointwise, then $f = \limsup_n f_n$. □

For a measurable $f : E \rightarrow \overline{\mathbb{R}}$ define its **positive** and **negative parts**

$$f^+ = \max\{f, 0\}, \quad f^- = \max\{-f, 0\}.$$

Both are non-negative measurable functions, at most one of them is non-zero at each point, and

$$f = f^+ - f^-, \quad |f| = f^+ + f^-.$$

2.3 Almost Everywhere**DEFINITION 2.8** *Almost everywhere*

Let $P(x)$ be a statement about points $x \in E$. We say that P holds μ -**almost everywhere** (a.e.) on E if there is $N \in \mathcal{F}$ with $\mu(N) = 0$ such that $P(x)$ holds for all $x \in E \setminus N$.

Note that only the set of points where P fails is required to be contained in a null set; that set itself need not be measurable. Typical uses are $f = g$ a.e., $f_n \rightarrow f$ a.e., and $f < \infty$ a.e.

EXAMPLE 2.9

(i) Consider

$$f(x) = \begin{cases} 2x, & x < 2, \\ 1, & x = 2, \\ x^3, & x > 2. \end{cases}$$

f is discontinuous only at $x = 2$ and $m(\{2\}) = 0$, so f is continuous almost everywhere on $\mathbb{R}$. But f is **not** almost everywhere equal to any continuous function: if g is continuous and $g = f$ a.e., then $g(x) = 2x$ on a dense subset of $(-\infty, 2)$ (the complement of a null set is dense), so by continuity $g(x) = 2x$ for all $x < 2$; likewise $g(x) = x^3$ for all $x > 2$. Then $g(2) = 4 \neq 1$, a contradiction.

(ii) The Dirichlet function $D = \mathbf{1}_{\mathbb{Q}}$ satisfies $D = 0$ a.e. (as $m(\mathbb{Q}) = 0$), so it is a.e. equal to the continuous function 0, yet D is discontinuous everywhere.

Thus “continuous almost everywhere” and “almost everywhere equal to a continuous function” are different notions, and neither implies the other.

PROPOSITION 2.10

Let $(X, \mathcal{F}, \mu)$ be complete (for example $(\mathbb{R}^n, \mathcal{L}, m)$).

- (i) If f is measurable and $g = f$ a.e., then g is measurable.
- (ii) If the f_n are measurable and $f_n \rightarrow f$ a.e., then f is measurable.

Proof. (i) Let N be a null set with $f = g$ on N^c . Then

$$\{g > a\} = (\{f > a\} \setminus N) \cup (\{g > a\} \cap N).$$

The first set lies in $\mathcal{F}$; the second is a subset of the null set N and lies in $\mathcal{F}$ by completeness. (ii) By Proposition 2.7, $h = \limsup_n f_n$ is measurable and $f = h$ a.e.; apply (i). $\square$

REMARK 2.11

Completeness cannot be dropped. By Article I, §4.5, the Cantor set C has a subset A that is not a Borel set. Then $\mathbf{1}_A = 0$ m -a.e., but $\mathbf{1}_A$ is not Borel measurable. This is one reason for preferring the complete space $(\mathbb{R}^n, \mathcal{L}, m)$. In a general measure space it suffices to remember that the integral (§3) does not distinguish functions that agree a.e.; hence a function that is undefined or equal to $\pm\infty$ on a null set can be modified there and treated as an everywhere-defined real-valued function.

2.4 Simple Functions

DEFINITION 2.12 *Simple function*

A measurable function $\varphi : X \rightarrow \mathbb{R}$ taking only finitely many real values is called **simple**. If $c_1, \dots, c_N$ are its distinct values and $A_k = \{\varphi = c_k\}$, then the $A_k \in \mathcal{F}$ are pairwise disjoint with union X , and

$$\varphi = \sum_{k=1}^N c_k \mathbf{1}_{A_k}.$$

This is the **canonical representation** of φ .

Every finite linear combination $\sum_{i=1}^n a_i \mathbf{1}_{E_i}$ ($a_i \in \mathbb{R}$, $E_i \in \mathcal{F}$) is simple, and conversely. Note that simple functions take **finite** real values and are therefore bounded; this point is crucial in Remark 3.7.

THEOREM 2.13 *Approximation by simple functions*

Let $f : X \rightarrow [0, +\infty]$ be measurable. There are simple functions φ_n with

$$0 \leq \varphi_1 \leq \varphi_2 \leq \dots \leq f, \quad \varphi_n(x) \rightarrow f(x) \quad (\forall x \in X),$$

and the convergence is uniform on every set on which f is bounded.

Proof. For $n \geq 1$ let

$$\varphi_n = \sum_{k=1}^{n2^n} \frac{k-1}{2^n} \mathbf{1}_{E_{n,k}} + n \mathbf{1}_{F_n}, \quad E_{n,k} = \left\{ \frac{k-1}{2^n} \leq f < \frac{k}{2^n} \right\}, \quad F_n = \{f \geq n\}.$$

These sets are measurable by Proposition 2.2, so φ_n is simple. Equivalently,

$$\varphi_n(x) = \min \{n, 2^{-n} \lfloor 2^n f(x) \rfloor\} \quad (\varphi_n(x) = n \text{ if } f(x) = +\infty).$$

Since $\lfloor 2s \rfloor \geq 2 \lfloor s \rfloor$, we have $2^{-(n+1)} \lfloor 2^{n+1} t \rfloor \geq 2^{-n} \lfloor 2^n t \rfloor$, whence $\varphi_n \leq \varphi_{n+1}$; clearly $\varphi_n \leq f$. If $f(x) = +\infty$, then $\varphi_n(x) = n \rightarrow +\infty$; if $f(x) < \infty$, then for $n \geq f(x)$,

$$0 \leq f(x) - \varphi_n(x) < 2^{-n}.$$

On $\{f \leq M\}$ this estimate holds for all $n \geq M$ simultaneously. $\square$

COROLLARY 2.14

If $f : X \rightarrow \overline{\mathbb{R}}$ is measurable, there are simple functions $\varphi_n \rightarrow f$ pointwise with $|\varphi_1| \leq |\varphi_2| \leq \dots \leq |f|$.

Proof. Apply Theorem 2.13 to f^+ and f^- to obtain $\psi_n \uparrow f^+$, $\chi_n \uparrow f^-$, and set $\varphi_n = \psi_n - \chi_n$. Since f^+ and f^- are never simultaneously non-zero, neither are ψ_n and χ_n , so $|\varphi_n| = \psi_n + \chi_n$ is increasing and bounded by $f^+ + f^- = |f|$. $\square$

2.5 Lusin's Theorem

Littlewood summarized the theory of functions of a real variable in three principles: every measurable set is “nearly” a finite union of intervals; every measurable function is “nearly” continuous; every pointwise convergent sequence of measurable functions is “nearly” uniformly convergent. The precise form of the second principle is Lusin's theorem. (The third is Egorov's theorem, which we do not discuss here.)

THEOREM 2.15 *Lusin*

Let $E \subseteq \mathbb{R}^n$ be Lebesgue measurable with $m(E) < \infty$, and let $f : E \rightarrow \overline{\mathbb{R}}$ be Lebesgue measurable and finite a.e. Then for every $\varepsilon > 0$ there is a compact set $F \subseteq E$ with

$$m(E \setminus F) < \varepsilon$$

such that the restriction $f|_F : F \rightarrow \mathbb{R}$ is continuous.

Proof. We use inner regularity (Article I, §4.6): if $A \in \mathcal{L}$ and $m(A) < \infty$, then for every $\eta > 0$ there is a compact $K \subseteq A$ with $m(A \setminus K) < \eta$.

Let $Z = \{|f| = \infty\}$, so $m(Z) = 0$, and put $E_0 = E \setminus Z$; on E_0 the function f is real-valued. Enumerate the open intervals with rational endpoints as $I_1, I_2, \dots$ and let

$$A_k = E_0 \cap f^{-1}(I_k), \quad B_k = E_0 \setminus A_k.$$

These are measurable sets of finite measure. Choose compact sets $K_k \subseteq A_k$ and $K'_k \subseteq B_k$ with

$$m(A_k \setminus K_k) < \frac{\varepsilon}{2^{k+2}}, \quad m(B_k \setminus K'_k) < \frac{\varepsilon}{2^{k+2}},$$

so that $m(E_0 \setminus (K_k \cup K'_k)) < \varepsilon/2^{k+1}$. Let

$$F = \bigcap_{k=1}^{\infty} (K_k \cup K'_k).$$

F is a closed subset of the compact set $K_1 \cup K'_1$, hence compact, and $F \subseteq E_0 \subseteq E$. Moreover

$$m(E \setminus F) \leq m(Z) + \sum_{k=1}^{\infty} m(E_0 \setminus (K_k \cup K'_k)) < \frac{\varepsilon}{2}.$$

It remains to show that $g = f|_F$ is continuous. Fix k . Since $K_k \subseteq A_k$, $K'_k \subseteq B_k$, $A_k \cap B_k = \emptyset$ and $F \subseteq K_k \cup K'_k$,

$$g^{-1}(I_k) = F \cap A_k = F \cap K_k = F \setminus K'_k = F \cap (\mathbb{R}^n \setminus K'_k),$$

which is relatively open in F . Every open subset of $\mathbb{R}$ is a union of some of the I_k , so the preimage under g of every open set is relatively open in F ; that is, g is continuous. $\square$

REMARK 2.16

(i) The conclusion is that the **restriction** $f|_F$ is continuous, not that f is continuous at the

points of F . For the Dirichlet function D and $E = [0, 1]$, take an open set $U \supseteq \mathbb{Q} \cap [0, 1]$ of total measure less than ε ; then $F = [0, 1] \setminus U$ is compact and $D|_F \equiv 0$ is continuous, although D is continuous at no point.

- (ii) The hypothesis $m(E) < \infty$ can be removed, with F closed: apply the theorem with $\varepsilon/2^k$ to $E \cap \{k-1 \leq |x| < k\}$ ($k \geq 1$); the union of the resulting compact sets is closed, since every bounded set meets only finitely many of them.
- (iii) Combined with the Tietze extension theorem, there is a continuous $g : \mathbb{R}^n \rightarrow \mathbb{R}$ with $g = f$ on F , so $m(\{x \in E : f(x) \neq g(x)\}) < \varepsilon$; if $|f| \leq M$, one may also require $|g| \leq M$. This is the precise meaning of “measurable functions can be approximated arbitrarily well by continuous functions”.
- (iv) More generally, Lusin’s theorem holds for Radon measures on locally compact Hausdorff spaces; see Rudin, *Real and Complex Analysis*, Theorem 2.24.

3 The Lebesgue Integral

Throughout this section $(X, \mathcal{F}, \mu)$ is fixed. The integral is defined in three steps: non-negative simple functions, non-negative measurable functions, general measurable functions.

3.1 Integrals of Non-negative Simple Functions

DEFINITION 3.1

Let $\varphi = \sum_{k=1}^N c_k \mathbf{1}_{A_k}$ be the canonical representation of a non-negative simple function ($c_k \geq 0$). Define

$$\int_X \varphi \, d\mu = \sum_{k=1}^N c_k \mu(A_k) \in [0, +\infty],$$

with the convention $0 \cdot \infty = 0$. For $E \in \mathcal{F}$ put $\int_E \varphi \, d\mu = \int_X \varphi \mathbf{1}_E \, d\mu$.

REMARK 3.2

The convention $0 \cdot \infty = 0$ ensures that a function which vanishes on a set of infinite measure gets no contribution from that set. For example, the zero function $0 = 0 \cdot \mathbf{1}_{\mathbb{R}}$ on $\mathbb{R}$ has integral $0 \cdot m(\mathbb{R}) = 0$. It does **not** mean that integrals over sets of infinite measure vanish: if some $c_k > 0$ has $\mu(A_k) = \infty$, the integral is $+\infty$.

LEMMA 3.3

If a non-negative simple function can be written as $\varphi = \sum_{i=1}^n a_i \mathbf{1}_{E_i}$ with $a_i \geq 0$ and $E_i \in \mathcal{F}$ (not necessarily disjoint), then

$$\int_X \varphi d\mu = \sum_{i=1}^n a_i \mu(E_i).$$

In particular, the integral does not depend on the representation.

Proof. For $\epsilon = (\epsilon_1, \dots, \epsilon_n) \in \{0, 1\}^n$ let $G_\epsilon = \bigcap_{i=1}^n E_i^{\epsilon_i}$, where $E^1 = E$ and $E^0 = E^c$. These “atoms” are pairwise disjoint, their union is X , and

$$E_i = \bigsqcup_{\epsilon: \epsilon_i=1} G_\epsilon.$$

On G_ϵ the function φ is constant, equal to $b_\epsilon = \sum_i \epsilon_i a_i$. By finite additivity,

$$\sum_{i=1}^n a_i \mu(E_i) = \sum_{i=1}^n a_i \sum_{\epsilon: \epsilon_i=1} \mu(G_\epsilon) = \sum_{\epsilon} b_\epsilon \mu(G_\epsilon),$$

where we have interchanged finite sums of terms in $[0, \infty]$. On the other hand, in the canonical representation $A_k = \bigsqcup_{\epsilon: b_\epsilon=c_k} G_\epsilon$, so $\sum_k c_k \mu(A_k)$ also equals $\sum_{\epsilon} b_\epsilon \mu(G_\epsilon)$. $\square$

PROPOSITION 3.4

Let φ, ψ be non-negative simple functions and $a, b \geq 0$.

- (i) $\int (a\varphi + b\psi) d\mu = a \int \varphi d\mu + b \int \psi d\mu$;
- (ii) if $\varphi \leq \psi$, then $\int \varphi d\mu \leq \int \psi d\mu$;
- (iii) $\nu(E) = \int_E \varphi d\mu$ is a measure on $\mathcal{F}$.

Proof. (i) If $\varphi = \sum c_k \mathbf{1}_{A_k}$ and $\psi = \sum d_j \mathbf{1}_{B_j}$, then $a\varphi + b\psi = \sum ac_k \mathbf{1}_{A_k} + \sum bd_j \mathbf{1}_{B_j}$; apply Lemma 3.3. (ii) $\psi - \varphi$ is a non-negative simple function (here we use that simple functions are finite-valued), so by (i), $\int \psi = \int \varphi + \int (\psi - \varphi) \geq \int \varphi$. (iii) $\varphi \mathbf{1}_E = \sum_k c_k \mathbf{1}_{A_k \cap E}$, so $\nu(E) = \sum_k c_k \mu(A_k \cap E)$. Each $E \mapsto \mu(A_k \cap E)$ is a measure, and a non-negative linear combination of measures is a measure (series of non-negative terms may be rearranged freely). $\square$

3.2 Integrals of Non-negative Measurable Functions

Let $L^+ = L^+(X, \mathcal{F})$ denote the set of measurable functions $f : X \rightarrow [0, +\infty]$.

DEFINITION 3.5

For $f \in L^+$ define

$$\int_X f d\mu = \sup \left\{ \int_X \varphi d\mu : \varphi \text{ simple, } 0 \leq \varphi \leq f \right\} \in [0, +\infty],$$

and for $E \in \mathcal{F}$ put $\int_E f d\mu = \int_X f \mathbf{1}_E d\mu$.

By Proposition 3.4(ii), if f is itself simple the supremum is attained at $\varphi = f$, so this definition agrees with the previous one.

PROPOSITION 3.6

Let $f, g \in L^+$ and $E, E' \in \mathcal{F}$.

- (i) If $f \leq g$, then $\int f \leq \int g$;
- (ii) for $c \in [0, \infty)$, $\int cf = c \int f$;
- (iii) if $E \subseteq E'$, then $\int_E f \leq \int_{E'} f$;
- (iv) if $\mu(E) = 0$, then $\int_E f = 0$.

Proof. (i) The supremum on the left is taken over a smaller set. (ii) For $c = 0$ both sides are 0; for $c > 0$, $\varphi \mapsto c\varphi$ is a bijection between $\{0 \leq \varphi \leq f\}$ and $\{0 \leq \varphi \leq cf\}$. (iii) Follows from $f\mathbf{1}_E \leq f\mathbf{1}_{E'}$ and (i). (iv) If φ is simple with $0 \leq \varphi \leq f\mathbf{1}_E$, then φ vanishes outside E ; in its canonical representation every A_k with $c_k > 0$ is contained in E and hence null, so $\int \varphi = 0$. $\square$

REMARK 3.7 On “upper integrals”

One might try to use

$$\inf \left\{ \int \varphi d\mu : \varphi \text{ simple, } \varphi \geq f \right\}$$

as an equivalent definition. In general this is **false**, because simple functions are bounded and take finitely many values:

- (i) On $[0, 1]$ let $f(x) = x^{-1/2}$ for $x > 0$ and $f(0) = 0$. Since f is unbounded, no simple $\varphi \geq f$ exists and the infimum is $\inf \emptyset = +\infty$, whereas $\int_{[0,1]} f d\mu = 2$ (Example 5.8).
- (ii) On $\mathbb{R}$ let $f(x) = e^{-|x|}$. A simple $\varphi \geq f > 0$ is everywhere positive, hence bounded below by its smallest value $c > 0$, so $\int \varphi \geq c \cdot m(\mathbb{R}) = +\infty$, whereas $\int_{\mathbb{R}} f d\mu = 2$.

The two quantities do agree when f is bounded and $\mu(X) < \infty$: by Theorem 2.13, $\varphi_n \leq f \leq \varphi_n + 2^{-n}$ for large n , so the infimum is at most $\int \varphi_n + 2^{-n}\mu(X)$; let $n \rightarrow \infty$.

3.3 The Monotone Convergence Theorem

The following theorem is the cornerstone of the whole theory; even the additivity of the integral is proved by means of it.

THEOREM 3.8 Monotone Convergence Theorem

Let $f_n \in L^+$ with $f_1 \leq f_2 \leq \dots$ pointwise, and $f = \lim_n f_n$. Then

$$\int_X f d\mu = \lim_{n \rightarrow \infty} \int_X f_n d\mu.$$

Proof. By Proposition 2.7, $f = \sup_n f_n \in L^+$. By monotonicity, $\int f_n$ increases and is bounded by $\int f$, so $\alpha = \lim_n \int f_n$ exists and $\alpha \leq \int f$.

For the reverse inequality, fix a simple function $0 \leq \varphi \leq f$ and $t \in (0, 1)$, and let

$$E_n = \{f_n \geq t\varphi\}.$$

If $\varphi = \sum_k c_k \mathbf{1}_{A_k}$, then $E_n = \bigcup_k (A_k \cap \{f_n \geq tc_k\})$ is measurable. The sets E_n increase and $\bigcup_n E_n = X$: if $\varphi(x) = 0$, then $x \in E_1$; if $\varphi(x) > 0$, then $t\varphi(x) < \varphi(x) \leq f(x)$, so $f_n(x) \geq t\varphi(x)$ for large n . Therefore

$$\int_X f_n d\mu \geq \int_{E_n} f_n d\mu \geq t \int_{E_n} \varphi d\mu = t \nu(E_n),$$

where $\nu(E) = \int_E \varphi d\mu$ is a measure (Proposition 3.4(iii)). By continuity from below (Article I, §2.3), $\nu(E_n) \rightarrow \nu(X) = \int \varphi$, so $\alpha \geq t \int \varphi$. Letting $t \rightarrow 1$ and taking the supremum over φ gives $\alpha \geq \int f$. $\square$

COROLLARY 3.9 Additivity

For $f, g \in L^+$, $\int (f + g) d\mu = \int f d\mu + \int g d\mu$.

Proof. By Theorem 2.13 choose simple $\varphi_n \uparrow f$ and $\psi_n \uparrow g$; then $\varphi_n + \psi_n \uparrow f + g$. By Proposition 3.4(i), $\int (\varphi_n + \psi_n) = \int \varphi_n + \int \psi_n$; apply the Monotone Convergence Theorem to all three sequences. $\square$

COROLLARY 3.10 Term-by-term integration

Let $f_k \in L^+$ ($k \geq 1$). Then

$$\int_X \sum_{k=1}^{\infty} f_k d\mu = \sum_{k=1}^{\infty} \int_X f_k d\mu.$$

Proof. Let $S_n = \sum_{k=1}^n f_k$. By Corollary 3.9 and induction, $\int S_n = \sum_{k=1}^n \int f_k$. Since S_n increases pointwise to $\sum_{k=1}^{\infty} f_k$, the Monotone Convergence Theorem gives the result as $n \rightarrow \infty$. $\square$

COROLLARY 3.11

For $f \in L^+$, $\nu(E) = \int_E f d\mu$ is a measure on $\mathcal{F}$, and $\mu(E) = 0$ implies $\nu(E) = 0$.

Proof. If $E = \bigsqcup_k E_k$, then $f \mathbf{1}_E = \sum_k f \mathbf{1}_{E_k}$, and Corollary 3.10 gives countable additivity; the second statement is Proposition 3.6(iv). $\square$

EXAMPLE 3.12

Let $\#$ be counting measure on $\mathbb{N}$ ($\#(A)$ is the number of elements of A , $\mathcal{F} = \mathcal{P}(\mathbb{N})$). Every $f : \mathbb{N} \rightarrow [0, \infty]$ can be written $f = \sum_j f(j) \mathbf{1}_{\{j\}}$, so by Corollary 3.10

$$\int_{\mathbb{N}} f d\# = \sum_{j=1}^{\infty} f(j).$$

Thus series of non-negative terms are special cases of integrals. Applying Corollary 3.10 to $f_k(j) = a_{jk} \geq 0$ shows that the order of summation of a non-negative double series may be

interchanged:

$$\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} a_{jk} = \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} a_{jk}.$$

3.4 The Markov and Chebyshev Inequalities

PROPOSITION 3.13 *Markov's inequality*

Let $f \in L^+$ and $\lambda \in (0, \infty)$. Then

$$\mu(\{f \geq \lambda\}) \leq \frac{1}{\lambda} \int_X f \, d\mu.$$

A fortiori the same holds for $\{f > \lambda\}$.

Proof. $\lambda \mathbf{1}_{\{f \geq \lambda\}} \leq f$; integrate. □

COROLLARY 3.14 *Chebyshev's inequality*

Let $f : X \rightarrow \mathbb{R}$ be measurable, $c \in \mathbb{R}$ and $\lambda > 0$. Then

$$\mu(\{|f - c| \geq \lambda\}) \leq \frac{1}{\lambda^2} \int_X |f - c|^2 \, d\mu.$$

Proof. $\{|f - c| \geq \lambda\} = \{|f - c|^2 \geq \lambda^2\}$; apply Markov's inequality to $|f - c|^2$. □

On a probability space ($\mu(X) = 1$), taking $c = \int f \, d\mu$ gives the familiar form $\mathbb{P}(|\xi - \mathbb{E}\xi| \geq \lambda) \leq \text{Var}(\xi)/\lambda^2$: the deviation of a random variable from its mean is controlled by its variance. The inequality of Proposition 3.13 is usually called Markov's inequality, although in the literature it is also often called Chebyshev's inequality.

COROLLARY 3.15

Let $f, g \in L^+$.

- (i) If $\int f < \infty$, then $f < \infty$ a.e.;
- (ii) $\int f = 0$ if and only if $f = 0$ a.e.;
- (iii) if $f = g$ a.e., then $\int f = \int g$;
- (iv) if $f_n \in L^+$ and $f_n \uparrow f$ a.e., then $\int f_n \rightarrow \int f$.

Proof. (i) $\{f = \infty\} \subseteq \{f \geq k\}$, so by Markov's inequality $\mu(\{f = \infty\}) \leq \frac{1}{k} \int f \rightarrow 0$. (ii) If $\int f = 0$, then $\mu(\{f \geq 1/k\}) \leq k \int f = 0$, and $\{f > 0\} = \bigcup_k \{f \geq 1/k\}$. Conversely, if $N = \{f > 0\}$ is null, then $\int f = \int_N f = 0$ (Proposition 3.6(iv)). (iii) Let N be a null set with $f = g$ on N^c . By Corollary 3.11,

$$\int f = \int_{N^c} f + \int_N f = \int_{N^c} g + 0 = \int g.$$

(iv) Let N be a null set such that $f_n \uparrow f$ on N^c . Apply the Monotone Convergence Theorem to $f_n \mathbf{1}_{N^c} \uparrow f \mathbf{1}_{N^c}$ and then use (iii). $\square$

3.5 Integrals of General Measurable Functions

DEFINITION 3.16 *Lebesgue integral*

Let $f : X \rightarrow \overline{\mathbb{R}}$ be measurable. If at least one of $\int f^+ d\mu$ and $\int f^- d\mu$ is finite, define

$$\int_X f d\mu = \int_X f^+ d\mu - \int_X f^- d\mu \in \overline{\mathbb{R}}.$$

If both are finite, f is called **integrable** (Lebesgue integrable). The set of integrable real-valued measurable functions is denoted $L^1(\mu) = L^1(X, \mathcal{F}, \mu)$. For $E \in \mathcal{F}$, $\int_E f d\mu = \int_X f \mathbf{1}_E d\mu$.

Since $|f| = f^+ + f^-$, Corollary 3.9 gives $\int |f| = \int f^+ + \int f^-$, so

$$f \text{ is integrable} \iff \int_X |f| d\mu < \infty.$$

REMARK 3.17

- (i) The Lebesgue integral is an **absolute** integral: f is integrable if and only if $|f|$ is. Hence there is no distinction between “integrable” and “absolutely integrable” in the Lebesgue theory; the difference between “convergent” and “absolutely convergent” only arises for improper Riemann integrals (§5.2).
- (ii) If f is integrable, it is finite a.e. by Corollary 3.15(i). Modifying f on a null set does not change its integral (Corollary 3.15(iii)), so integrable functions may always be regarded as real-valued. In Article III, L^1 will be understood as a space of equivalence classes modulo a.e. equality, with norm $\|f\|_1 = \int |f| d\mu$.
- (iii) A complex-valued $f = u + iv$ is integrable if u and v are, and then $\int f = \int u + i \int v$.

EXAMPLE 3.18

- (i) The Dirichlet function satisfies $D = 0$ a.e. on $[0, 1]$, so $\int_{[0,1]} D dm = 0$, even though $D \notin \mathcal{R}[0, 1]$.
- (ii) On $(0, 1]$, $1/x \geq 2^{k-1}$ on $(2^{-k}, 2^{-k+1}]$, an interval of measure 2^{-k} . By Corollary 3.11,

$$\int_{(0,1]} \frac{dx}{x} = \sum_{k=1}^{\infty} \int_{(2^{-k}, 2^{-k+1}]} \frac{dx}{x} \geq \sum_{k=1}^{\infty} \frac{1}{2} = +\infty.$$

The integral exists (and equals $+\infty$), but $1/x$ is not integrable.

- (iii) On $[-1, 1] \setminus \{0\}$ let $f(x) = 1/x$. By (ii) and symmetry, $\int f^+ = \int f^- = +\infty$, so $\int f dm$ is **undefined**, even though f is odd and all its symmetric truncations have integral zero.

3.6 Basic Properties

THEOREM 3.19

Let $f, g \in L^1(\mu)$ and $a, b \in \mathbb{R}$.

- (i) (Linearity) $af + bg \in L^1(\mu)$ and $\int (af + bg) = a \int f + b \int g$.
- (ii) (Monotonicity) If $f \leq g$ a.e., then $\int f \leq \int g$. In particular, $f = g$ a.e. implies $\int f = \int g$.
- (iii) $|\int f d\mu| \leq \int |f| d\mu$.
- (iv) If $E \in \mathcal{F}$ and $\mu(E) = 0$, then $\int_E f = 0$.
- (v) (Countable additivity over the domain) If $E_1, E_2, \dots \in \mathcal{F}$ are **pairwise disjoint** and $E = \bigsqcup_k E_k$, then

$$\int_E f d\mu = \sum_{k=1}^{\infty} \int_{E_k} f d\mu,$$

and the series converges absolutely.

- (vi) If $\mu(E) < \infty$ and $|f| \leq M$ a.e. on E , then $|\int_E f| \leq M\mu(E)$.

Proof. (i) By $|af + bg| \leq |a||f| + |b||g|$, Proposition 3.6 and Corollary 3.9, $\int |af + bg| \leq |a| \int |f| + |b| \int |g| < \infty$. Scalars: for $a \geq 0$, $(af)^\pm = af^\pm$; for $a < 0$, $(af)^+ = |a|f^-$ and $(af)^- = |a|f^+$; in both cases Proposition 3.6(ii) gives $\int af = a \int f$. Sums: let $h = f + g$. From

$$h^+ - h^- = f^+ - f^- + g^+ - g^-$$

we get $h^+ + f^- + g^- = h^- + f^+ + g^+$, an identity between functions in L^+ . By Corollary 3.9,

$$\int h^+ + \int f^- + \int g^- = \int h^- + \int f^+ + \int g^+,$$

and since all terms are finite we may rearrange to get $\int h = \int f + \int g$.

- (ii) $(g - f)^- = 0$ a.e., so $\int (g - f)^- = 0$ by Corollary 3.15(ii); hence $\int (g - f) = \int (g - f)^+ \geq 0$, and we conclude by (i).

- (iii) $|\int f| = |\int f^+ - \int f^-| \leq \int f^+ + \int f^- = \int |f|$.

- (iv) $f^\pm \mathbf{1}_E = 0$ a.e.

- (v) Apply Corollary 3.11 to f^+ and to f^- to obtain two convergent series of non-negative terms, and subtract. Absolute convergence follows from $\sum_k |\int_{E_k} f| \leq \sum_k \int_{E_k} |f| = \int_E |f| < \infty$.

- (vi) $-M\mathbf{1}_E \leq f\mathbf{1}_E \leq M\mathbf{1}_E$ a.e.; apply (ii). □

REMARK 3.20

Linearity has to be proved directly from the definition of the integral (via the Monotone Convergence Theorem); it cannot be deduced from “ L^1 is a vector space”, since the closure of L^1 under addition itself depends on (i). Also, “pairwise disjoint” in (v) means $E_i \cap E_j = \emptyset$ for $i \neq j$, not $\bigcap_k E_k = \emptyset$; the latter is a much weaker condition and does not imply additivity.

4 Convergence Theorems

We want to know when

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X \lim_{n \rightarrow \infty} f_n d\mu ?$$

Consider first three typical counterexamples in $(\mathbb{R}, \mathcal{L}, m)$. Each converges pointwise to 0, yet every integral equals 1:

- (i) **escape by translation:** $f_n = \mathbf{1}_{[n, n+1]}$;
- (ii) **escape by spreading:** $f_n = \frac{1}{n} \mathbf{1}_{[0, n]}$ (which even converges to 0 uniformly);
- (iii) **concentration:** $f_n = n \mathbf{1}_{(0, 1/n)}$.

In all three cases $\int \lim f_n = 0 < 1 = \lim \int f_n$: “mass” can be lost in the limit. Fatou’s lemma below says that, for non-negative functions, mass can only be lost, never created.

4.1 The Monotone Convergence Theorem Revisited

Combining Theorem 3.8 with Corollary 3.15(iv):

THEOREM 4.1 *Monotone Convergence Theorem, a.e. version*

Let f_n, f be measurable, $0 \leq f_1 \leq f_2 \leq \dots$ a.e., and $f_n \rightarrow f$ a.e. Then $\int f_n d\mu \uparrow \int f d\mu$.

REMARK 4.2

Neither non-negativity nor monotonicity can be dropped.

- (i) **Decreasing** sequences fail: $f_n = \mathbf{1}_{[n, \infty)} \downarrow 0$, but $\int f_n = +\infty$ for all n .
- (ii) **Increasing** sequences without a lower bound fail: $f_n = -\mathbf{1}_{[n, \infty)} \uparrow 0$, but $\int f_n = -\infty$ for all n .

To handle signed functions we first prove a lemma on “integrable perturbations”.

LEMMA 4.3

Let $h : X \rightarrow \overline{\mathbb{R}}$ be measurable with $\int h^- < \infty$, and let $k \in L^1(\mu)$. Then $\int (h + k)$ is defined and

$$\int (h + k) d\mu = \int h d\mu + \int k d\mu.$$

Proof. k is finite a.e., so $h + k$ is defined a.e. If $\int h^+ < \infty$, then $h \in L^1$ and the claim is Theorem 3.19(i). Suppose $\int h^+ = \infty$. Since $(h + k)^- \leq h^- + k^-$, $\int (h + k)^- < \infty$. Moreover

$$h^+ = h + h^- \leq (h + k)^+ + |k| + h^-,$$

so by Corollary 3.9, $\infty = \int h^+ \leq \int (h + k)^+ + \int |k| + \int h^-$, whence $\int (h + k)^+ = \infty$. Both sides therefore equal $+\infty$. $\square$

COROLLARY 4.4 *Monotone convergence with an integrable bound*

- (i) If $f_n \uparrow f$ a.e. and $\int f_1^- < \infty$, then $\int f_n \uparrow \int f$.
(ii) If $f_n \downarrow f$ a.e. and $\int f_1^+ < \infty$, then $\int f_n \downarrow \int f$.

Proof. (i) f_1^- is integrable, hence finite a.e. Let $h_n = f_n + f_1^-$. Then $h_n \geq f_1 + f_1^- \geq 0$ a.e. and $h_n \uparrow f + f_1^-$ a.e.; by Theorem 4.1, $\int h_n \rightarrow \int (f + f_1^-)$. Since $f_n^- \leq f_1^-$ and $f^- \leq f_1^-$, Lemma 4.3 with $h = f_n$ (or f) and $k = f_1^-$ gives $\int h_n = \int f_n + \int f_1^-$ and $\int (f + f_1^-) = \int f + \int f_1^-$. Cancel the finite number $\int f_1^-$. (ii) Apply (i) to $-f_n$. $\square$

4.2 Fatou's Lemma**THEOREM 4.5** *Fatou's lemma*

Let $f_n : X \rightarrow [0, +\infty]$ be measurable. Then

$$\int_X \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int_X f_n d\mu.$$

Proof. Let $g_k = \inf_{n \geq k} f_n$. As the infimum of **countably** many measurable functions it is measurable by Proposition 2.7; moreover $0 \leq g_1 \leq g_2 \leq \dots$ and $g_k \uparrow \liminf_n f_n$. Since $g_k \leq f_n$ for every $n \geq k$,

$$\int g_k \leq \inf_{n \geq k} \int f_n.$$

Applying the Monotone Convergence Theorem to (g_k) ,

$$\int \liminf_{n \rightarrow \infty} f_n = \lim_{k \rightarrow \infty} \int g_k \leq \lim_{k \rightarrow \infty} \inf_{n \geq k} \int f_n = \liminf_{n \rightarrow \infty} \int f_n.$$

 $\square$

The three examples at the beginning of this section show that the inequality may be strict. Since the Monotone Convergence Theorem only holds for non-negative functions (or functions with an integrable lower bound), the non-negativity hypothesis is indispensable.

COROLLARY 4.6 *Fatou's lemma with an integrable lower bound*

Let $g \in L^1(\mu)$ and let f_n be measurable with $f_n \geq g$ a.e. for all n . Then

$$\int \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int f_n d\mu.$$

Dually, if $f_n \leq g$ a.e., then $\limsup_n \int f_n \leq \int \limsup_n f_n$.

Proof. g is finite a.e. Put $h_n = f_n - g \geq 0$ a.e.; then $\liminf_n h_n = \liminf_n f_n - g$ a.e. By Lemma 4.3 (with $h = h_n$ or $h = \liminf_n h_n$, and $k = g$), $\int f_n = \int h_n + \int g$ and $\int \liminf_n f_n = \int \liminf_n h_n + \int g$. Apply Fatou's lemma to h_n (after modification on a null set to make them non-negative, which does

not affect the integrals) and cancel the finite number $\int g$. For the dual statement apply the first part to $-f_n \geq -g$. $\square$

REMARK 4.7

The lower bound g need only be integrable and may take negative values; if $g \geq 0$ is required, the statement is just Fatou's lemma itself. Without an integrable lower bound the conclusion may fail, even under pointwise convergence: on $(0, 1)$ let $f_n = -n\mathbf{1}_{(0, 1/n)}$. Then $f_n \rightarrow 0$ pointwise and $\int f_n = -1$, whereas $\int \liminf f_n = 0 > -1$.

4.3 The Dominated Convergence Theorem

THEOREM 4.8 *Lebesgue's Dominated Convergence Theorem*

Let f_n, f be measurable with $f_n \rightarrow f$ a.e., and suppose there is $g \in L^1(\mu)$ such that for all n

$$|f_n| \leq g \quad \text{a.e.}$$

Then $f_n, f \in L^1(\mu)$,

$$\lim_{n \rightarrow \infty} \int_X |f_n - f| d\mu = 0, \quad \text{and hence} \quad \lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu.$$

Proof. Passing to the limit, $|f| \leq g$ a.e., so f_n and f are integrable (after modification on a null set we may assume they are finite everywhere). Also $|f_n - f| \leq 2g$ a.e., so $h_n = 2g - |f_n - f| \geq 0$ a.e. and $h_n \rightarrow 2g$ a.e. By Fatou's lemma and linearity,

$$\int 2g \leq \liminf_{n \rightarrow \infty} \int (2g - |f_n - f|) = \int 2g - \limsup_{n \rightarrow \infty} \int |f_n - f|.$$

Since $\int 2g < \infty$, $\limsup_n \int |f_n - f| \leq 0$. Finally, by Theorem 3.19(iii), $|\int f_n - \int f| \leq \int |f_n - f| \rightarrow 0$. $\square$

REMARK 4.9

- (i) Both the Monotone and the Dominated Convergence Theorems only require convergence almost everywhere, and neither contains the other: the former needs no dominating function and allows $\int f = +\infty$; the latter needs no monotonicity but requires a common integrable dominating function.
- (ii) Domination cannot be dropped. None of the three examples at the beginning of this section admits an integrable dominating function; for $f_n = n\mathbf{1}_{(0, 1/n)}$, for instance, $\sup_n f_n(x) \geq \frac{1}{x} - 1$, which is not integrable on $(0, 1)$.
- (iii) Domination is sufficient but not necessary. On a finite measure space, a sequence of integrable functions converging in measure converges in L^1 if and only if it is uniformly integrable; this is the Vitali convergence theorem, treated in the supplementary article on uniform integrability.

COROLLARY 4.10 *Bounded Convergence Theorem*

Let $\mu(X) < \infty$, $|f_n| \leq M$ a.e. and $f_n \rightarrow f$ a.e. Then $\int f_n \rightarrow \int f$.

Proof. The constant M is integrable on a finite measure space; apply the Dominated Convergence Theorem. $\square$

COROLLARY 4.11 *Term-by-term integration of series*

Let the f_k be measurable with $\sum_{k=1}^{\infty} \int |f_k| d\mu < \infty$. Then $\sum_k f_k(x)$ converges absolutely for almost every x , its sum is integrable, and

$$\int_X \sum_{k=1}^{\infty} f_k d\mu = \sum_{k=1}^{\infty} \int_X f_k d\mu.$$

Proof. Let $G = \sum_k |f_k|$. By Corollary 3.10, $\int G = \sum_k \int |f_k| < \infty$, so $G < \infty$ a.e., and at such points the series converges absolutely. The partial sums satisfy $|\sum_{k \leq n} f_k| \leq G$; apply the Dominated Convergence Theorem and linearity. $\square$

EXAMPLE 4.12

We compute $\lim_{n \rightarrow \infty} \int_0^n \left(1 + \frac{x}{n}\right)^n e^{-2x} dx$. Let $f_n(x) = (1 + x/n)^n e^{-2x} \mathbf{1}_{[0,n]}(x)$. From $\log(1+t) \leq t$ we get $(1 + x/n)^n \leq e^x$, hence

$$0 \leq f_n(x) \leq e^{-x},$$

and e^{-x} is integrable on $[0, \infty)$ with integral 1 (Theorem 5.6). For fixed $x \geq 0$, $f_n(x) \rightarrow e^x e^{-2x} = e^{-x}$. By the Dominated Convergence Theorem the limit equals $\int_{[0, \infty)} e^{-x} dx = 1$.

4.4 Integrals Depending on a Parameter**THEOREM 4.13**

Let $I \subseteq \mathbb{R}$ be an open interval and $f : X \times I \rightarrow \mathbb{R}$ with $f(\cdot, t) \in L^1(\mu)$ for each $t \in I$. Put

$$F(t) = \int_X f(x, t) d\mu(x).$$

- (i) If $t \mapsto f(x, t)$ is continuous at t_0 for every x , and there is $g \in L^1(\mu)$ with $|f(x, t)| \leq g(x)$ for all x, t , then F is continuous at t_0 .
- (ii) If $t \mapsto f(x, t)$ is differentiable on I for every x , and there is $g \in L^1(\mu)$ with $|\partial_t f(x, t)| \leq g(x)$ for all x, t , then F is differentiable on I and

$$F'(t) = \int_X \partial_t f(x, t) d\mu(x).$$

Proof. (i) Let $t_n \rightarrow t_0$. Then $f(\cdot, t_n) \rightarrow f(\cdot, t_0)$ pointwise with domination by g , so $F(t_n) \rightarrow F(t_0)$ by the Dominated Convergence Theorem. By the sequential characterization of limits, F is continuous at t_0 .

(ii) Fix $t \in I$ and $t_n \rightarrow t$ with $t_n \neq t$, and let

$$h_n(x) = \frac{f(x, t_n) - f(x, t)}{t_n - t}.$$

Each h_n is measurable and $h_n(x) \rightarrow \partial_t f(x, t)$, so $\partial_t f(\cdot, t)$ is measurable. By the mean value theorem $h_n(x) = \partial_t f(x, \tau_n)$ for some $\tau_n = \tau_n(x)$ between t and t_n , so $|h_n| \leq g$. By the Dominated Convergence Theorem,

$$\frac{F(t_n) - F(t)}{t_n - t} = \int_X h_n d\mu \rightarrow \int_X \partial_t f(x, t) d\mu(x).$$

Since the sequence $t_n \rightarrow t$ was arbitrary, the claim follows. $\square$

5 Riemann and Lebesgue Integrals

5.1 The Proper Riemann Integral

THEOREM 5.1

Let $f : [a, b] \rightarrow \mathbb{R}$ with $f \in \mathcal{R}[a, b]$. Then f is Lebesgue measurable, $f \in L^1([a, b], m)$, and

$$\int_{[a, b]} f dm = \int_a^b f(x) dx.$$

THEOREM 5.2 *Lebesgue's criterion*

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and let D_f be its set of discontinuities. Then

$$f \in \mathcal{R}[a, b] \iff m(D_f) = 0;$$

that is, a bounded function is Riemann integrable if and only if it is continuous almost everywhere.

Both theorems are proved with the same construction. Let $|f| \leq M$. For $k \geq 0$ let P_k be the dyadic partition

$$x_j^{(k)} = a + j \frac{b-a}{2^k}, \quad j = 0, 1, \dots, 2^k.$$

Then $P_{k+1} \supseteq P_k$ and $\|P_k\| = (b-a)2^{-k} \rightarrow 0$. For $x \in (x_{j-1}^{(k)}, x_j^{(k)})$ define

$$l_k(x) = \inf_{[x_{j-1}^{(k)}, x_j^{(k)}]} f, \quad u_k(x) = \sup_{[x_{j-1}^{(k)}, x_j^{(k)}]} f,$$

and put $l_k(a) = u_k(a) = f(a)$. The l_k, u_k are Borel measurable simple functions, $l_k \leq f \leq u_k$, and (points being null sets)

$$\int_{[a, b]} l_k dm = L(f, P_k), \quad \int_{[a, b]} u_k dm = U(f, P_k).$$

Since P_{k+1} refines P_k and an infimum over a smaller interval can only be larger, $l_k \leq l_{k+1}$; similarly $u_{k+1} \leq u_k$. Let $l = \lim_k l_k$ and $u = \lim_k u_k$. These are Borel measurable, $|l|, |u| \leq M$, and $l \leq f \leq u$. By the Bounded Convergence Theorem (Corollary 4.10),

$$\int_{[a,b]} l \, dm = \lim_{k \rightarrow \infty} L(f, P_k), \quad \int_{[a,b]} u \, dm = \lim_{k \rightarrow \infty} U(f, P_k). \quad (*)$$

Proof of Theorem 5.1. By Theorem 1.8, $U(f, P_k)$ and $L(f, P_k)$ both converge to $\int_a^b f(x) \, dx$. By (*), $\int (u - l) \, dm = 0$ with $u - l \geq 0$, so $u = l$ a.e. (Corollary 3.15(ii)). Since $l \leq f \leq u$, $f = l$ a.e. As l is Borel measurable and $(\mathbb{R}, \mathcal{L}, m)$ is complete, f is Lebesgue measurable by Proposition 2.10. Being bounded on a set of finite measure, f is integrable, and

$$\int_{[a,b]} f \, dm = \int_{[a,b]} l \, dm = \lim_{k \rightarrow \infty} L(f, P_k) = \int_a^b f(x) \, dx.$$

□

LEMMA 5.3

Let $Z = \bigcup_k P_k$ (a countable set). For $x \in [a, b] \setminus Z$, f is continuous at x if and only if $u(x) = l(x)$.

Proof. Since $x \notin Z$, for each k the point x lies in the interior of exactly one subinterval $J_k(x)$ of P_k , and $l_k(x) = \inf_{J_k(x)} f$, $u_k(x) = \sup_{J_k(x)} f$.

($\Rightarrow$) Given $\varepsilon > 0$, choose $\delta > 0$ with $|f(y) - f(x)| < \varepsilon$ whenever $|y - x| < \delta$. If $\|P_k\| < \delta$, then $J_k(x) \subseteq (x - \delta, x + \delta)$, hence $u_k(x) - l_k(x) \leq 2\varepsilon$. Thus $u(x) - l(x) \leq 2\varepsilon$ for every ε .

($\Leftarrow$) Given $\varepsilon > 0$, since $u_k(x) - l_k(x) \downarrow u(x) - l(x) = 0$, choose k with $u_k(x) - l_k(x) < \varepsilon$. As x is an interior point of $J_k(x)$, there is $\delta > 0$ with $(x - \delta, x + \delta) \subseteq J_k(x)$. For such y , both $f(y)$ and $f(x)$ lie in $[l_k(x), u_k(x)]$, so $|f(y) - f(x)| < \varepsilon$. □

Proof of Theorem 5.2. ($\Rightarrow$) By the proof of Theorem 5.1, $u = l$ a.e. By Lemma 5.3, $D_f \subseteq Z \cup \{u \neq l\}$, a union of two null sets.

($\Leftarrow$) If $m(D_f) = 0$, then by Lemma 5.3 the set $\{u \neq l\} \subseteq D_f \cup Z$ is null, so $\int (u - l) \, dm = 0$. By (*), $U(f, P_k) - L(f, P_k) \rightarrow 0$, and Theorem 1.8 gives $f \in \mathcal{R}[a, b]$. □

REMARK 5.4

- (i) Theorem 5.1 shows that the Lebesgue integral is an extension of the proper Riemann integral: $\mathcal{R}[a, b] \subseteq L^1([a, b])$ with the same integral. The boundedness assumption in Lebesgue's criterion is necessary: unbounded functions are never Riemann integrable (Proposition 1.4).
- (ii) The Dirichlet function is discontinuous everywhere, hence not Riemann integrable, but it is Lebesgue integrable with integral 0. Thomae's function ($1/q$ at a rational p/q in lowest terms, 0 at irrationals) is discontinuous exactly at the rationals, a countable set, and is therefore Riemann integrable.
- (iii) Arzelà's theorem from Remark 1.12 is now immediate: if $f_n, f \in \mathcal{R}[a, b]$, $|f_n| \leq M$

and $f_n \rightarrow f$ pointwise, then by Theorem 5.1 and the Bounded Convergence Theorem $\int_a^b f_n = \int_{[a,b]} f_n dm \rightarrow \int_{[a,b]} f dm = \int_a^b f$. In Example 1.10, the Bounded Convergence Theorem likewise gives $\int_{[0,1]} f_n dm \rightarrow \int_{[0,1]} D dm = 0$, a limit that cannot even be expressed in the Riemann framework.

- (iv) The sequence of Example 1.13 converges in $L^1([0, 1])$ to $x^{-1/2}$: by the Monotone Convergence Theorem, $\int_{[0,1]} |f_n - x^{-1/2}| dm = \int (x^{-1/2} - f_n) dm \rightarrow 0$.

5.2 Improper Integrals

DEFINITION 5.5 *Improper integrals*

- (i) (Unbounded interval) Let $f : [a, \infty) \rightarrow \mathbb{R}$ with $f \in \mathcal{R}[a, b]$ for every $b > a$. If the limit

$$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx$$

exists and is finite, the improper integral is said to converge.

- (ii) (Unbounded integrand) Let $f : (a, b] \rightarrow \mathbb{R}$ be unbounded near a with $f \in \mathcal{R}[c, b]$ for every $c \in (a, b)$. If $\lim_{c \rightarrow a^+} \int_c^b f(x) dx$ exists and is finite, the improper integral $\int_a^b f(x) dx$ is said to converge.

If the corresponding improper integral of $|f|$ converges, the improper integral is **absolutely convergent**; if it converges but not absolutely, it is **conditionally convergent**.

A singular point of the second kind is a point where the function is **unbounded**, not merely a point of discontinuity: a bounded function with finitely many jump discontinuities on a bounded interval is Riemann integrable in the proper sense (Theorem 5.2) and needs no improper integral. Note also that if $f \in \mathcal{R}[a, b]$, then $f^\pm, |f| \in \mathcal{R}[a, b]$, since their discontinuities are discontinuities of f .

THEOREM 5.6

Let $f : [a, \infty) \rightarrow \mathbb{R}$ be Riemann integrable on every $[a, b]$. Then f is Lebesgue measurable on $[a, \infty)$, and:

- (i) if $f \geq 0$, then $\int_{[a,\infty)} f dm = \lim_{b \rightarrow \infty} \int_a^b f(x) dx \in [0, +\infty]$;
- (ii) $f \in L^1([a, \infty))$ if and only if $\int_a^\infty |f(x)| dx$ converges, and then $\int_{[a,\infty)} f dm = \int_a^\infty f(x) dx$;
- (iii) if $\int_a^\infty f(x) dx$ converges conditionally, then $\int_{[a,\infty)} f^+ dm = \int_{[a,\infty)} f^- dm = +\infty$, so

$\int_{[a,\infty)} f \, dm$ is undefined; nevertheless

$$\int_a^\infty f(x) \, dx = \lim_{b \rightarrow \infty} \int_{[a,b]} f \, dm.$$

Entirely analogous statements hold for improper integrals of unbounded integrands.

Proof. By Theorem 5.1, f is Lebesgue measurable on each $[a, a+j]$, and $\{f > c\} = \bigcup_j \{x \in [a, a+j] : f(x) > c\}$, so f is measurable on $[a, \infty)$.

(i) Let $b_n \uparrow \infty$. Then $f \mathbf{1}_{[a,b_n]} \uparrow f \mathbf{1}_{[a,\infty)}$, and by the Monotone Convergence Theorem and Theorem 5.1,

$$\int_{[a,\infty)} f \, dm = \lim_{n \rightarrow \infty} \int_a^{b_n} f(x) \, dx.$$

Since $b \mapsto \int_a^b f$ is non-decreasing, the limit along sequences is the limit as $b \rightarrow \infty$.

(ii) Applying (i) to $|f|$ gives $\int_{[a,\infty)} |f| \, dm = \lim_{b \rightarrow \infty} \int_a^b |f|$, which proves the first assertion. If $|f|$ is integrable, then for every $b_n \uparrow \infty$ we have $f \mathbf{1}_{[a,b_n]} \rightarrow f \mathbf{1}_{[a,\infty)}$ with domination by $|f|$, so by the Dominated Convergence Theorem

$$\int_a^{b_n} f(x) \, dx = \int_{[a,b_n]} f \, dm \rightarrow \int_{[a,\infty)} f \, dm.$$

(iii) By (ii), $\int |f| \, dm = \infty$, so at least one of $\int f^+$ and $\int f^-$ is $+\infty$. If only one were infinite, say $\int f^- \, dm < \infty = \int f^+ \, dm$, then by (i) applied to f^+ ,

$$\int_a^b f(x) \, dx = \int_{[a,b]} f^+ \, dm - \int_{[a,b]} f^- \, dm \geq \int_{[a,b]} f^+ \, dm - \int_{[a,\infty)} f^- \, dm \rightarrow +\infty,$$

contradicting the convergence of the improper integral; the other case is symmetric. The last identity is just Theorem 5.1 together with the definition of the improper integral. $\square$

EXAMPLE 5.7 $\int_0^\infty \frac{\sin x}{x} \, dx$

Let $f(x) = \frac{\sin x}{x}$ for $x > 0$ and $f(0) = 1$; f is continuous.

(a) f is not Lebesgue integrable. On $[k\pi, (k+1)\pi]$ we have $|f(x)| \geq \frac{|\sin x|}{(k+1)\pi}$, and $\int_{k\pi}^{(k+1)\pi} |\sin x| \, dx = 2$, so

$$\int_0^{N\pi} |f(x)| \, dx \geq \frac{2}{\pi} \sum_{k=1}^N \frac{1}{k} \rightarrow \infty.$$

(b) The improper integral converges to $\pi/2$. For $t \geq 0$ and $0 < R < S$ let $g(x) = e^{-tx}/x$, which is positive and decreasing on $(0, \infty)$. Integrating by parts,

$$\int_R^S g(x) \sin x \, dx = [-g(x) \cos x]_R^S + \int_R^S g'(x) \cos x \, dx,$$

and $\int_R^S |g'| = g(R) - g(S)$, so

$$\left| \int_R^S e^{-tx} \frac{\sin x}{x} dx \right| \leq g(R) + g(S) + g(R) - g(S) = 2g(R) \leq \frac{2}{R}. \quad (**)$$

By the Cauchy criterion the improper integral $I(t) = \int_0^\infty e^{-tx} f(x) dx$ converges for every $t \geq 0$, and for all $t \geq 0$,

$$\left| I(t) - \int_0^R e^{-tx} f(x) dx \right| \leq \frac{2}{R}.$$

For $t > 0$, $|e^{-tx} f(x)| \leq e^{-tx}$ is integrable, so $I(t) = \int_{[0,\infty)} e^{-tx} f dm$ by Theorem 5.6(ii). For $t \geq t_0 > 0$, $|\partial_t(e^{-tx} f(x))| = e^{-t_0 x} |\sin x| \leq e^{-t_0 x}$, and Theorem 4.13(ii) gives

$$I'(t) = - \int_0^\infty e^{-tx} \sin x dx = -\frac{1}{1+t^2} \quad (t > 0),$$

the last integral being computed from the antiderivative $-e^{-tx}(t \sin x + \cos x)/(1+t^2)$. Hence $I(t) = C - \arctan t$. Moreover $|I(t)| \leq \int_0^\infty e^{-tx} dx = 1/t \rightarrow 0$ as $t \rightarrow \infty$, so $C = \pi/2$.

Finally we show that I is right-continuous at $t = 0$. For every $R > 0$,

$$|I(t) - I(0)| \leq \left| \int_{[0,R]} (e^{-tx} - 1)f(x) dx \right| + \frac{4}{R}.$$

On $[0, R]$, $|(e^{-tx} - 1)f(x)| \leq 1$ and the integrand tends to 0 pointwise as $t \rightarrow 0^+$, so the first term tends to 0 by the Bounded Convergence Theorem. Hence $\limsup_{t \rightarrow 0^+} |I(t) - I(0)| \leq 4/R$, and letting $R \rightarrow \infty$,

$$\int_0^\infty \frac{\sin x}{x} dx = I(0) = \lim_{t \rightarrow 0^+} \left(\frac{\pi}{2} - \arctan t \right) = \frac{\pi}{2}.$$

(c) Conclusion. This is a conditionally convergent improper integral. By Theorem 5.6(iii), $\int_{[0,\infty)} f^+ dm = \int_{[0,\infty)} f^- dm = +\infty$ and the Lebesgue integral $\int_{[0,\infty)} f dm$ is undefined; all one can say is that

$$\lim_{R \rightarrow \infty} \int_{[0,R]} \frac{\sin x}{x} dm = \frac{\pi}{2}.$$

EXAMPLE 5.8 Unbounded integrands

(i) $\int_0^1 x^{-1/2} dx = \lim_{c \rightarrow 0^+} (2 - 2\sqrt{c}) = 2$. By the analogue of Theorem 5.6(i) for unbounded integrands, $\int_{(0,1]} x^{-1/2} dm = 2$, so $x^{-1/2} \in L^1((0, 1])$.

(ii) Consider $\int_0^1 \frac{1}{x} \sin \frac{1}{x} dx$. Substituting $x = 1/s$,

$$\int_c^1 \frac{1}{x} \sin \frac{1}{x} dx = \int_1^{1/c} \frac{\sin s}{s} ds, \quad \int_c^1 \left| \frac{1}{x} \sin \frac{1}{x} \right| dx = \int_1^{1/c} \frac{|\sin s|}{s} ds.$$

By Example 5.7, as $c \rightarrow 0^+$ the former converges while the latter diverges. So this improper integral converges conditionally, and the integrand is not in $L^1((0, 1])$.

REMARK 5.9

To summarize the relation between the Riemann and Lebesgue integrals:

- (i) proper Riemann integrable $\Rightarrow$ Lebesgue integrable, with the same integral (Theorem 5.1);
- (ii) absolutely convergent improper integral $\iff$ Lebesgue integrable, with the same integral (Theorem 5.6(ii));
- (iii) a conditionally convergent improper integral corresponds to no Lebesgue integral and can only be expressed as a limit of Lebesgue integrals (Theorem 5.6(iii));
- (iv) conversely, a Lebesgue integrable function (such as the Dirichlet function) may be neither properly nor improperly Riemann integrable.

Hence “every Riemann integrable function is Lebesgue integrable” is correct for proper integrals; caution is needed only for conditionally convergent improper integrals.

Summary

Starting from the Riemann integral, we saw that its main weakness concerns limits: limits of integrable functions need not be integrable, interchanging limits and integrals requires strong hypotheses such as uniform convergence, and the space of integrable functions with the integral distance is not complete.

Lebesgue’s remedy is to partition the range rather than the domain. This requires measurable functions, i.e. functions with $\{f > a\} \in \mathcal{F}$. Measurable functions are closed under algebraic operations and countable limits, can be approximated monotonically by simple functions, and (Lusin’s theorem) are nearly continuous.

The integral is defined in three steps:

$$\begin{aligned}
 \text{simple functions:} & \quad \int \sum_k c_k \mathbf{1}_{A_k} d\mu = \sum_k c_k \mu(A_k), \\
 \text{non-negative functions:} & \quad \int f d\mu = \sup_{0 \leq \varphi \leq f} \int \varphi d\mu, \\
 \text{general functions:} & \quad \int f d\mu = \int f^+ d\mu - \int f^- d\mu.
 \end{aligned}$$

The Monotone Convergence Theorem is the cornerstone of the theory; from it we obtained additivity, term-by-term integration, Fatou’s lemma and finally the Dominated Convergence Theorem:

$$f_n \rightarrow f \text{ a.e., } |f_n| \leq g \in L^1 \implies \int |f_n - f| d\mu \rightarrow 0.$$

Finally, the Lebesgue integral does extend the proper Riemann integral and characterizes Riemann integrability as almost-everywhere continuity, while its relation to improper integrals is governed by absolute convergence.

The next article (Article III) builds on this to study normed spaces and L^p spaces, proving the inequalities of Hölder and Minkowski and the completeness of L^p (the Riesz–Fischer theorem). The latter answers precisely the incompleteness of the space of Riemann integrable functions observed in §1.3.

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